1. IC collection, aimed gather: a larger same-session A/A (2026-10-29)

Study: `research/ic_aimed_gather_20261023/aa2_20261029/` · **Protocol:** `PROTOCOL.md`, committed and pushed (`5d726412b`) before the first run · **Arms:** the first session's binaries, by `sha256` (`runs/arms.sha256`): main `d34b9b390` and the change `6bc18857b` (merged in #2448) · **Workload:** `ic price` on rule v3 `k0n53/T01.params.json`, curve `icv1-f2m53-tm56619371-dac20a85` · **Host:** one x86-64 cloud VM, Intel Xeon @ 2.10 GHz, 4 vCPU, AVX-512, kernel `6.18.44-fc-v185`, no hardware PMU (`runs/host.json`).

1.1. Question

The first session (2026-10-23) measured the collect phase 1.111× faster over 30 paired rounds ([1.082, 1.147] with the corrected bootstrap below; published as [1.065, 1.160]), but its A/A had 5 pairs, with an upper end of 1.087. The rule declared before measuring needs the A/B interval's lower end above 1 and above the A/A interval's upper end, so the verdict was **not established**. This session measures the noise floor with as many rounds as the effect, in the same session.

1.2. Design

42 rounds. Each round runs baseline `a`, baseline `q` and candidate `c`, in the order given by cycling through all six permutations, so each label runs in each position seven times. Per round: A/B = `a/c`, A/A = `a/q`. Every process ran under `isolated_bench --cpus 3`, `RAYON_NUM_THREADS=1`. The statistic and rule are the first session's: the median of per-round ratios, a 95% percentile bootstrap over rounds (2,000 resamples, a fixed LCG, seed 20261029).

1.3. Result (measured)

metric	A/B median	A/B 95%	rounds > 1	A/A 95%	verdict
collect phase	1.093	1.068 – 1.136	38 / 42	0.977 – 1.025	**established**
whole set-up	1.042	1.020 – 1.082	29 / 42	0.946 – 1.016	marginal (+0.005), not claimed

Correction. The bootstrap as first run (in `summarise.sh` and the first session's `../summarise.sh`) stepped its LCG in jq's floating point; the rounded states are almost always even, so 99.6% of resample draws fell on odd-numbered rounds. The independent validator found it. The intervals above use the same constants, seed and resample count with the step done exactly (`correct_bootstrap.sh`, `summary-corrected.json`); `CORRECTION.md` lists the as-run intervals beside them. As run, the collect interval was 1.090 – 1.160 against an A/A upper end of 1.030, and the whole set-up was not established.

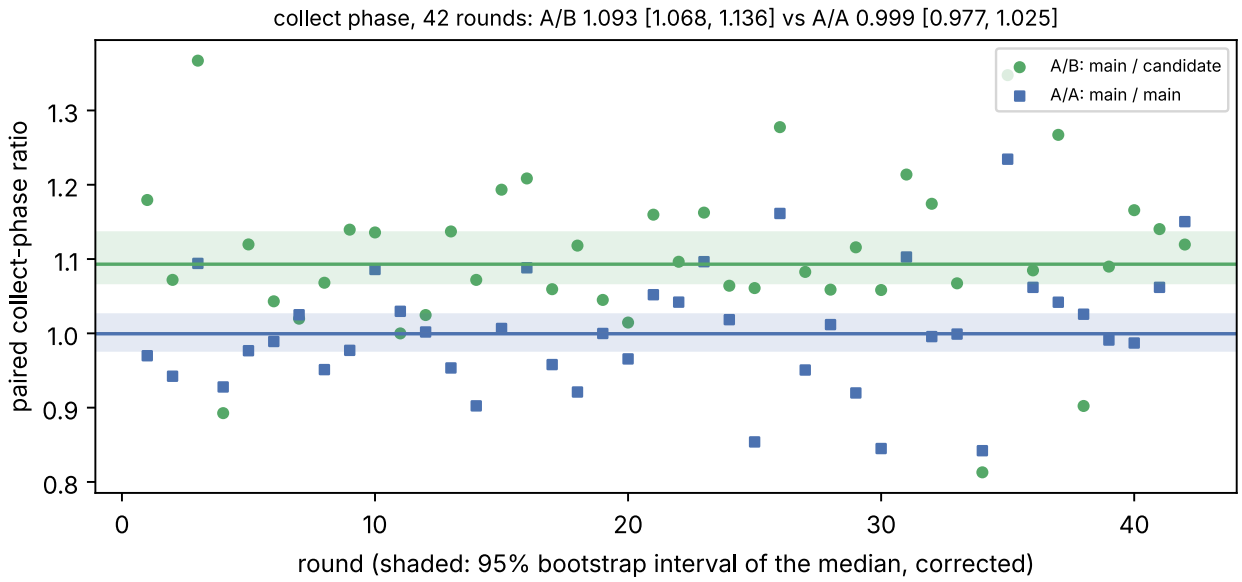


Figure 1: Per-round collect-phase ratios, A/B (green) and A/A (blue), with each median and its corrected 95% interval (`runs/`, `summary.json`, `summary-corrected.json`).

- **Collect phase: established.** The A/B lower end, 1.068, clears 1 and the A/A upper end, 1.025, by 0.042. It also held under the as-run generator, and under 500 of 500 independent seeds (validator). Median collect time: 230.3 ms (main `a`) and 235.2 ms (main `q`) against 212.5 ms (candidate).
- **Whole set-up: marginal, not claimed.** Corrected, 1.020 clears the A/A's 1.016 by 0.005; as run, it did not. The generator was fixed after the data were seen, so this verdict is not claimed. The set-up includes phases the change does not touch.
- **Isolation:** 126 processes, none contended, none failed, no round dropped. 71 launches were refused for CPU pressure and retried; a refusal measures nothing.
- **Identity:** all 126 recovered `518509155375` with `all_verified: true`; an independent replay matched every non-timing field in both arms.

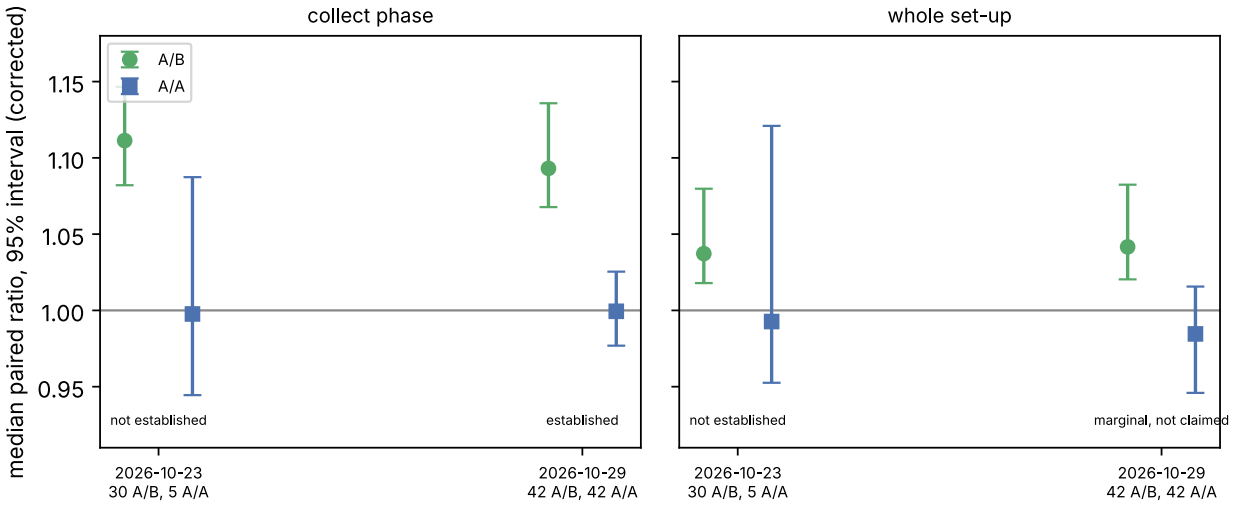


Figure 2: Both sessions' A/B and A/A intervals, corrected. The first session's A/B effect is the same size; its 5-pair A/A was too wide to rule on, and its verdicts do not change with the correction.

1.4. What this changes, and what it does not

- **Class: engineering**, a native component speed-up. On the measured params every counted operation is identical between arms; identity was checked on these params only.
- **The board's row is not re-priced here.** Rule v7 times S, so a faster collect phase should eventually lower the `icv1-f2m53-tm56619371-dac20a85` one-target row (IC 8.146 against rho 1.159 S, 7.03×). Re-pricing it needs rule v7's 64-target protocol and is not claimed.
- **Not pooled** with the first session: different kernel, so a different host state. The first session's records are unchanged.
- **Not measured:** other sizes' native time, other hosts, IPC (no PMU). The run was local, not on Iso Lab: the reachable EKS paths run ecbench specs only, and this build-against-build A/A needs each round's three processes on one host (`PROTOCOL.md`, "Route").